

Exact-Relation Coding: A Preregistered Test of Whether Discovered Fixed-Width GF(2)/Affine Structure Yields a Composed Lossless-Compression Advantage

Preregistered empirical study · inconclusive for the full corpus, clean negative within the achieved coverage
frozen experiment state: git tag `v1.1-final` · repository: <https://github.com/Xeloraa/deductive-coding>
every quantity traceable to `results/ledger.json`

Author(s): withheld for review — repository: <https://github.com/Xeloraa/deductive-coding> **Frozen experiment state:** git tag `v1.1-final` (this PDF was generated from exactly that commit). **Numerical provenance:** every quantity in this paper is generated from `results/ledger.json` (122 experiment records) by `scripts/regen_tables.py`; every inline figure additionally carries a source marker naming the experiment record and field it is drawn from, verified against the ledger by `scripts/check_paper_numbers.py`.

Scope in one sentence. The negative result concerns *fixed-width, axis-aligned, GF(2) linear relations* (plus an integer-affine variant used only for tabular controls, and one bounded bit-phase extension); it is **not** a test of "algebraic redundancy" in general.

Abstract

Lossless compressors model a probability for the next symbol and code a residual. We investigate an alternative pre-pass: automatically *discover* an exact algebraic relation the data satisfies, transmit a fully counted description of that relation together with the independent symbols, and let the decoder reconstruct the determined symbols. We instantiate the pre-pass as a fixed-width GF(2) column-basis codec (with an integer-affine variant used only for tabular controls), define a *composed deduction gain* $G = \min_c |c(x)| - \min_c |c(D(x))|$ over a fixed set of stock compressors c and the accounted container $D(x)$, and preregister a per-file success threshold before running any natural-corpus experiment. On data engineered to contain an exact GF(2) linear code the pre-pass removes about half of the size the six stock baselines achieve (1 MiB code: $G = 523,726$ bytes, $G_{\text{pct}} = +49.95\%$); the gain scales with the number of rows while relation description stays near-constant, and two context-mixing compressors (paq8l, paq8px v216) do not close it. We then test whether comparable exploitable *exact GF(2)* structure transfers to public natural corpora.

The preregistered natural corpus is the 12 Silesia members (whole), enwik8, one SDRBench scientific `float32` field, and one telemetry text file. On the available 8 GiB machine we ran **8 of the 12 Silesia members whole** (≤ 10.2 MB); the other four members and whole enwik8 / SDRBench / UCI exceed memory and were run only at 256 KiB. **Strictly by the preregistration the hypothesis outcome is therefore inconclusive with respect to the complete corpus** (the whole-file list is not finished). *Within the achieved coverage* — 8 whole Silesia members, all 12 members plus enwik8 plus six `float32` fields plus the telemetry file at 256 KiB, and a bounded bit-phase-offset extension — the result is a **clean layered negative**: no non-prior-art file produces a composed gain meeting the threshold; the largest G_{pct} on any natural file is **+0.09 % ($G_{\text{pct}} = 0.0009$)**, on a passthrough container — a sub-0.1 % downstream-compressor sensitivity to an 18-byte header, not deduction — and the offset extension reproduces the phase-0 composed gain to the byte on every file. Functional-dependency and per-record-CRC32 cases reproduce known techniques and are labelled as such.

The mechanism is not novel (syndrome source coding; grammar compression; recent program-synthesis compression). Our contribution is the accounting-complete codec with an independent shared-nothing verifier, the composed-gain evaluation discipline with a preregistered threshold, and the empirical finding that blind fixed-width axis-aligned GF(2) relation discovery does not yield a composed

advantage on the natural corpora and detectors tested. We do **not** claim natural data contains no exact algebraic redundancy, nor that the general approach can never help.

Keywords: lossless compression, algebraic redundancy, GF(2) linear codes, functional dependencies, preregistration, negative result, reproducibility.

1. Introduction

Redundancy in a byte string is either *statistically predictable* — a good model assigns the next symbol high probability — or *exactly determined* — some relation fixes it given other symbols. General-purpose compressors exploit the first. Whether it is ever worthwhile to explicitly discover and transmit the second, on arbitrary data, after paying for the description, is an old question that is periodically revisited.

It carries a known trap. A "win" measured against the payload alone, or against the transformed container *before* a downstream compressor, routinely evaporates once the container is itself compressed, because the stock compressor already captures the same redundancy given a more convenient byte layout. Any honest test must therefore (i) count every metadata bit inside the container and (ii) compare *after* a strong downstream compressor. We make that comparison the primary quantity and fix it, together with a success threshold, in a git-locked preregistration written before the natural-corpus experiments.

1.1 Research questions

We separate five layers and never let one stand in for another; "we found a relation" (RQ-A) is never reported as a compression result.

- **RQ-A — does exact structure exist?** Does x satisfy an exact GF(2)/affine relation at a tried bit width and phase?
- **RQ-B — can we discover and represent it?** Is the relation verified on *every* row and encoded into a well-formed, fully accounted container?
- **RQ-C — does it reduce total representation cost?** Is the container strictly smaller than both a never-worse passthrough and the best stock compressor on the raw bytes, *before* any downstream compressor?
- **RQ-D — does the reduction survive a strong downstream compressor?** Is the composed gain $G > 0$?
- **RQ-E — does it occur naturally at meaningful scale?** Across the preregistered natural corpus, does any non-prior-art file clear the fixed threshold?

RQ-A $\rightarrow$ RQ-B $\rightarrow$ RQ-C $\rightarrow$ RQ-D is a strict per-file funnel; RQ-E is the corpus-level aggregate.

1.2 Contributions

1. A GF(2) column-basis lossless codec with an integer-affine variant, a bit-exact accounting ledger whose finaliser refuses to emit a container whose category bits do not sum to its byte length, and a structural never-worse passthrough fallback (§3).
2. The *composed deduction gain* metric and a preregistered per-file threshold, as an evaluation discipline for pre-pass compression claims (§2, §4).
3. An independent, shared-nothing decoder and end-to-end composed round-trip verification (§5.4).
4. A control battery that makes a negative result *informative*: planted positive, i.i.d./shuffled/near-relation nulls, a corruption sweep, a representation-change false-positive estimate, a metadata-cost check, a composition-order check, and a non-aligned-period scope control (§6).

5. A preregistered empirical finding: on 8 whole Silesia members, 20 further files at 256 KiB, and a bounded bit-phase-offset extension, blind fixed-width axis-aligned GF(2) relation discovery yields **no** composed advantage meeting the threshold, robustly to bit phase (§8) — with the whole-file list for the full preregistered corpus not yet complete (§8.4).

The mechanism itself is not claimed as novel; see §7.

2. Problem definition and the composed metric

Let x be the input byte string and B a fixed set of lossless stock compressors (§5.2). Let E be a lossless deductive encoder with $E^{-1}(E(x)) = x$, and write $D(x) = E(x)$ for the *container*. E emits the byte-smaller of a relation container (§3) and *passthrough* (`magic · version · kind · len64 · crc32 · x`, all whole-byte fields, so $|passthrough(x)| = |x| + 18$), hence $|D(x)| \leq |x| + 18$ always.

```
raw_best(x)      = min over c in B, c ran of |c(x)|
composed_best(x) = min over c in B, c ran of |c(D(x))|
G(x)             = raw_best(x) - composed_best(x)      (signed bytes)
G_pct(x)         = G(x) / raw_best(x)
```

$G(x) > 0$ is the event the hypothesis needs. A companion *raw gain* $raw_best(x) - |D(x)|$ (no downstream compressor) is reported to expose the case where a raw win vanishes under composition. When a compressor fails on one side only (e.g. `xz` runs out of memory on a large container but not on the raw bytes), a *matched-codec gain* over the intersection of compressors that ran on both sides is also recorded; a positive is reportable only if the matched gain also clears the threshold.

3. Method

3.1 GF(2) fixed-width column basis

For a bit width $w \in \{8, 16, 32, 64, 128, 256\}$, the bits of x are reshaped to an $\lceil 8|x|/w \rceil \times w$ matrix; the $< w$ trailing bits are carried verbatim (`leftover`). Gaussian elimination over GF(2) on packed 64-bit rows computes the leftmost column basis: the pivot columns are the unique leftmost independent set, and every free column is expressed as an XOR of pivot columns. The map is checked against the *original* matrix before use — a discovery bug raises an error rather than producing a wrong container. Homogeneous and affine (`[1 | A]`, the all-ones column synthesised at decode and never transmitted) variants are both tried, and the smallest fully accounted container over all (`w, variant`) is kept, else *passthrough*.

3.2 Integer affine functional dependencies

For byte strings that parse as an `int64` table, candidate relations $z = \sum a_i x_i + b$ are solved on a few rows and verified with exact Python integers on *every* row; coefficients are zig-zag variants, counted. This variant is used only for synthetic functional-dependency controls; the natural-corpus experiments use §3.1 exclusively.

3.3 Bit-phase-offset extension

The detectors of §3.1–3.2 are *axis-aligned* and reshape from bit 0. As the one bounded detector-broadening attempt required by the preregistration's kill criterion, we add a phase sweep: for each width w , also reshape starting at every bit phase $p \in \{0, \dots, w-1\}$ (a coarse p -subset for $w \in \{128, 256\}$), carrying the skipped p bits as a counted `prefix` field. This tests whether a phase-0 negative is a *framing* artifact — a genuine w -periodic linear relation beginning mid-byte is invisible to phase-0 reshaping and recovered at the right phase (verified: a planted width-24 code with 8 junk bits prepended is found at w

= 24, $p = 8$ and missed at phase 0). It is bounded — $\sum w$ extra reshapes, no search over nonlinear forms — and does not attempt a learned permutation.

3.4 Container and bit accounting

The relation container is (LSB-first, then byte padding):

```
magic[4] version[8] kind[8]
n_rows[32] n_cols[32] n_payload_pivots[32]
flags[8] (bit0 affine, bit1 ones-is-pivot, bit2 has-prefix)
original_nbytes[64] leftover_nbits[32] leftover bits
prefix_nbits[32] + prefix bits (only if flags bit2)
pivot_mask[n_cols]
per free column, in index order: coefficients[n_payload_pivots (+1 if affine)]
pivot payload: n_rows · n_payload_pivots bits (row-major, pivot-column order)
crc32(x)[32]
padding to a whole byte
```

Every emitted bit is attributed by `AccountedWriter` to exactly one of {header, relation, payload, leftover, prefix, crc, framing}. The finaliser raises unless

$$8 \cdot |D(x)| = \sum \text{category bits}, \quad \text{framing} = (-\sum \text{others}) \bmod 8 \in [0, 7].$$

A container with an unaccounted bit cannot be emitted, and a mis-typed category name still counts toward the total (it lands in an overflow bucket), so the invariant cannot be defeated by a typo. `crc32(x)` is written and re-checked on decode; it is not needed for reconstruction, so it makes the deductive container *larger* (conservative for the hypothesis) and is identical work on the passthrough side (neutral for $\mathbb{G}$).

3.5 Decoding

Decode reads the fields in the same order, rebuilds the matrix by `pivot_bits · coeffsT` over $\text{GF}(2)$ (a `uint8` matmul; the low bit of the wrapping sum is the parity), reattaches the prefix and leftover, truncates to `original_nbytes`, and checks the CRC and length. The encoder is deterministic (asserted byte-identical across re-runs for 18 frozen cases).

4. Preregistration

`docs/preregistration.md` is git-locked; the commit that adds it is an ancestor of every natural-corpus result. It fixes, before any natural-corpus number existed:

- **Meaningful positive, per file:** $G_{\text{pct}} \geq 0.05$ **and** $G \geq 1024$ bytes **and** exact round-trip **and** the container is a real deduction ($n_{\text{relations}} \geq 1$, not passthrough) **and** the file is not a format-trick, derived-column, or checksum case. The 5 % floor is the midpoint of the 3–10 % band a compressor-generation change typically achieves; it is a relevance threshold, not a measurement-precision limit.
- **Hypothesis outcome:** *positive* iff ≥ 1 non-prior-art corpus meets the per-file threshold and reproduces within $\pm 10\%$; *negative* iff none does while the positive, null, and prior-art controls all behave as designed; *inconclusive* if a control fails or resource limits prevent running enough of the corpus list.
- **Kill criterion (three conjuncts):** negative on the full corpus list; **and** one bounded detector-broadening attempt moves no natural file above threshold; **and** the strongest executed context mixer does not itself close the planted- $\text{GF}(2)$ gap. Its status is tracked in `docs/kill_criterion_status.md`.

Post-lock changes are limited to engineering that cannot move a reported size (codec vectorisation guarded by byte-exact equivalence tests, result-file schema additions, plotting). Baselines, presets, accounting rules, and thresholds are frozen.

5. Experimental protocol

5.1 Datasets and corpus coverage

The preregistered natural corpus is: the 12 Silesia members (whole files); enwik8 (first 10^8 bytes); ≥ 1 SDRBench scientific `f1oat32` field; and the UCI Individual Household Electric Power Consumption text. The scientific and telemetry corpora were named in the lock, before any loader output was inspected, because they are the canonical public datasets for "exact cross-field structure is plausible here" — not because they produced a favourable number.

id	obtained	SHA-256 pinned	run as
<code>silesia_{dickens,xml,ooffice,reymont,sao,x-ray,mr,osdb}</code>	<code>silesia.zip</code> (Mahoney mirror)	yes	whole (5.3–10.2 MB)
<code>silesia_{samba,nci,webster,mozilla}</code>	same	yes	256 KiB prefix only
<code>enwik8</code>	<code>enwik8.zip</code>	yes	256 KiB prefix only
<code>sdrbench_exaalt2869440_{vx,vy,vz,xx,yy,zz}.f32</code>	SDRBench EXAALT bundle (Globus HTTPS), tar-verified	yes	256 KiB prefix (fields are 11.5 MB whole)
<code>uci_household_power_text</code>	UCI ML Repository #235	yes	256 KiB prefix only

Every corpus is pinned by SHA-256 in `results/corpus_manifest.json`; a changed hash aborts the run. No corpus has been dropped for being unfavourable; the four Silesia members not run whole are the *largest*, not the most inconvenient (§8). Prefixes are labelled as such and live in a separate result phase; they are never presented as whole-file results.

Train/test separation. The discovery step reads only the file it encodes; there is no learned model, no shared parameters across files, and no held-out tuning set. The only quantities fixed across all files are the six bit widths, the six stock compressors, and the preregistered threshold. Synthetic data is generated from documented integer seeds.

5.2 Baselines

`B = {gzip -9 (mtime 0), zlib -9, bzip2 -9, xz -9 (FORMAT_XZ), zstd -19, brotli -11}`, all deterministic; versions recorded per record (`zstandard` 0.25.0, `brotli` 1.2.0, `stdlib` otherwise; Python 3.13.6). A compressor that raises `MemoryError` or times out is recorded unavailable with the exception and excluded from the `min`; the available set is reported per record.

Two context-mixing compressors — `paq8l (-3, -8)` and `paq8px v216 (-4, -8)` — are applied **only** to the planted-GF(2) positive control, to test whether the mechanism's advantage there is a genuine blind spot of the strongest statistical compressors we could run. `cmix` (needs 20–32 GiB, no memory level) and `nncp` (needs a GPU) could not be run on the available hardware; the paper writes "the strongest context-mixing compressors we could run", never "state of the art".

5.3 Independent verification

`verification/independent_verify.py` shares no code with the encoder or the main decoder: its own bit reader, its own container parse, a plain-Python XOR / integer reconstruction, and an independent re-derivation of the accounting from parsed field sizes. It checks the full chain `raw → encode → container → compress → decompress → independent-decode → raw`, asserting SHA-256 equality and the $8 \cdot |D| = \sum \text{categories}$ invariant on the independently parsed field sizes. It reconstructs all four container kinds in a self-test; the CI suite exercises it over 21 parametrised cases (offsets 0–31, all container kinds, the composed compress/decompress chain, and a phase-shifted planted code it must recover); and it was run once on the whole 10 MB `silesia_dickens` container. Separately, the *primary* decoder's composed round-trip $E^{-1}(c^{-1}(c(D(x)))) = x$ is checked for every stock compressor on every one of the 122 ledger records — 0 failures.

5.4 Reproducibility harness

`scripts/reproduce.py` runs, in order: `pytest` (636 tests on the frozen state: byte-equivalence, randomised property tests, independent verifier), the verifier self-test, the controls, the natural-corpus slice sweep, the offset-extension sweep, the legacy phase experiments, then `build_ledger.py → regen_tables.py → make_figures.py → check_paper_numbers.py → independent_verify.py --ledger`. Every step returns 0 on the reported state. Every experiment JSON records dataset SHA-256, seeds, config, the seven accounting categories, container size, round-trip and composed round-trip status, every baseline (name, bytes, seconds, available), every composition size, `G`, `G_pct`, the raw gain, a UTC timestamp, the git commit (with a `-dirty` marker if the tree was not clean), and machine info.

6. Controls

All gate assertions pass (`experiments/controls/run.py`; full numbers in `paper/results_tables.md`).

control	construction	result
positive: planted GF(2), 3 seeds	random info + exact parity columns	<code>G_pct</code> +48.31 %, +49.5 %, +49.7 %; round-trip exact
null: i.i.d. bits / shuffled planted / 1-flip near-relation	—	passthrough; $ G \leq 19$ bytes
corruption sweep, flip fraction $\phi \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 5 \cdot 10^{-2}\}$	planted code, $\phi \cdot n_{\text{rows}} \cdot n_{\text{parity}}$ parity cells flipped	<code>G</code> decreases monotonically 16 213 → 13 238 → 1 402 → -18 → -18; never below -64
representation-change null	40 i.i.d. inputs through the full width × variant <code>min</code>	0 containers beat passthrough; 0 composed gains > 64 bytes
metadata-cost	null input; planted input	null overhead = exactly 18 bytes; planted container bits = $\sum \text{categories}$ to the bit
composition-order	planted container through <code>decompress(compress(.))</code> for five codecs	primary decoder inverts every time
non-aligned period, <code>w = 48</code>	exact 48-bit-period linear code	default widths → passthrough (<code>G -18</code>); adding width 48 → 24 relations, round-trip exact
prior art: affine FD / per-record CRC32	derived column / CRC records	<code>G</code> = +37 880 / +16 209 bytes, labelled FD elimination / checksum inversion; not counted toward RQ-E

The corruption sweep and the non-aligned-period control together bound the scope of a negative: the method degrades gracefully as exact structure is broken, and it is blind to exact structure whose period

is not among the tried widths (visible once the width is added).

7. Related work and novelty

The general idea — discover exact structure, transmit a reconstruction recipe, and beat a general-purpose compressor after counting the recipe — is not new, and we claim no novelty for it. Bracketed keys refer to §14.

- **Syndrome source coding** [Ancheta 1976]. Treat the source as an error pattern and transmit its syndrome Hx under a linear code; a coset-leader source decompresses exactly, and with no side information the coder is an entropy coder. Compressing a single source by exploiting linear-code membership is classical. What is not standard is *discovering* the code from one input with its description cost charged on the same channel — a thin distinction, and the axis on which our codec differs from classical syndrome coding.
- **Grammar-based compression** [Nevill-Manning & Witten 1997; Larsson & Moffat 2000; Kieffer & Yang 2000]. The string is the unique yield of a grammar whose description cost is counted; structure is concatenative rather than linear-algebraic.
- **Functional-dependency / derived-column elimination** [US 8,150,888; US 8,700,579; Liu et al. 2017 (TICC); Liu et al. 2024 (Corra); US 11,562,085]. Detect that a column is an exact function of others, drop it, and store the function; extensions store exceptions for inexact dependencies. This space is thoroughly occupied for relational tables. Our affine-FD and per-record-CRC32 results are labelled as reproductions of it and are excluded from RQ-E.
- **Format-aware recompression** [Schnaader, *Precomp*; preflate; packJPG]. Invert a known embedded codec and drop reconstructible checksums; our PNG/ZIP/SQLite behaviour is the predicted format-awareness trap, not a counterexample.
- **Program-synthesis compression** [Shi et al. 2026 (Brevis)]. Synthesise a self-contained DSL program that reconstructs a tensor bit-exactly; reports a real composed win on neural-network checkpoints — a favourable domain dense with repeats, low rank and quantisation grids, searched with a rich language guided by a learned prior. A strong recent instance of the general idea; it does not address blind fixed-width GF(2) discovery on arbitrary bytes.
- **Low GF(2)-rank / binary matrix factorisation** [Fomin et al. 2018]. A change of basis to expose binary rank deficiency; NP-hard in general, and the *exact* case is the special case our column basis computes at a fixed width.
- **Invariant mining** [Ernst et al. 2007 (Daikon)]. Templated dynamic discovery of equalities and affine relations among program variables; not a compressor, and it reports *likely* invariants (false positives expected), whereas we accept a relation only after verifying it on every row.

Our contribution, stated narrowly. (i) An accounting-complete, never-worse fixed-width GF(2) (plus tabular-affine) relation codec whose emitted bits are exhaustively categorised, with an independent shared-nothing verifier and end-to-end composed round-trip checks; (ii) the composed deduction-gain metric with a git-locked preregistered per-file threshold, offered as an evaluation discipline for pre-pass compression claims; (iii) a preregistered empirical measurement — with a bounded bit-phase-offset extension — of whether blind fixed-width axis-aligned GF(2) relation discovery produces a composed advantage on public natural corpora, finding that it does not, within the coverage tested. We do not claim any of (i)–(iii) is "first" in a stronger sense than this.

8. Results

8.1 Mechanism validation (planted GF(2))

On random GF(2) linear codes — k information columns and k exact parity columns — the pre-pass behaves exactly as designed. The relation description is a constant of the code (1,088 bits for $k = 32$; 4,224 bits for $k = 64$), independent of the number of rows, so the composed gain grows with the row count: G_{pct} rises from +32.31 % at 256 bytes to +49.95 % at 1 MiB (Fig. 1; Table B.2). Every case round-trips, and the independent verifier agrees.

Context mixers do not close this gap. On a 10 KiB code (seed 902), the best stock compressor leaves +4 949 bytes recoverable by deduction; paq8l -8 compresses the raw bytes to 10 321 and the container to 5 347 (a mixer-relative gap of +4 974), and paq8px v216 -8 to 10 261 and 5 288 (+4 973). Raising the mixer memory level changes the gap by < 6 bytes. The planted-GF(2) composed gain is therefore not an artifact of weak baselines; it is a blind spot of these statistical compressors on this bitstream. This is a statement about those compressors on that input, not a theorem about statistical models in general.

8.2 Natural corpora (RQ-C, RQ-D, RQ-E)

Whole files (8 of 12 Silesia members). No file meets the threshold (Table B.3). Three members admit GF(2) relations at a tried width but the container is *larger* than the best stock compressor's output, so RQ-C already fails; composition then makes it worse:

member	bytes	kind	rels	raw_best	G (bytes)	G_pct
dickens	10,192,446	GF2	25	2,799,520	-2,645,016	-94.48 %
x-ray	8,474,240	GF2	58	4,051,112	-1,907,747	-47.09 %
xml	5,345,280	GF2	21	430,390	-1,299,038	-301.83 %

The relations found on text/XML are the "high bit of a mostly-ASCII byte is zero" bit-plane and similar; brotli and bzip2 already exploit this from the raw byte order. The other five members admit no exact relation at any tried width and fall to passthrough, with $|G| \leq 6\,408$ bytes ($|G_{\text{pct}}| \leq 0.15$ %), i.e. downstream-compressor sensitivity to an 18-byte header, not deduction.

256 KiB files (20). Same pattern: 0 meaningful positives (Table B.4). Where GF(2) finds relations (float32 fields, XML, text, telemetry) the composed gain is large-negative (-4.8 % to -302 %); where it does not, $|G_{\text{pct}}| < 0.3$ %. The single largest positive G_{pct} on any natural-corpus record — whole files, 256 KiB files, and the offset extension together — is +0.09 % ($G_{\text{pct}} = 0.0009$), on the 256 KiB prefix of `silesia_sao`, whose container is *passthrough* ($n_{\text{relations}} = 0$). This is downstream-compressor sensitivity to the 18-byte passthrough header, not deduction; it is two orders of magnitude below the 5 % floor and is excluded from the hypothesis in any case because it is not a real deduction. (The +49.95 % maximum over all 122 records is the planted positive control, §8.1.)

Bit-phase-offset extension (20 files). For every file the offset-search composed gain equals the phase-0 value **to the byte** (Table B.5); no bit phase improves on phase 0 by more than header perturbation, and 0 files cross the threshold. The axis-aligned negative is robust to bit phase; it is not an artifact of reshaping from bit 0. This is the one bounded detector-broadening attempt the kill criterion requires, and it does not rescue the hypothesis.

8.3 Prior-art sanity cases

An integer affine derived column ($C = 3A + 5B + 7$, 8 192 rows) yields $G = +37\,880$ bytes and a per-record IEEE CRC32 set yields $G = +16\,209$ bytes (affine) / +15 701 bytes (homogeneous). These are labelled reproductions of functional-dependency elimination and checksum inversion respectively and are **not** counted toward RQ-E.

8.4 Verdict under the preregistered protocol

Coverage against the full preregistered corpus.

preregistered item	preregistered "run as"	actually run
Silesia — 12 members	whole files	8 whole (dickens, xml, ooffice, reymont, sao, x-ray, mr, osdb, ≤ 10.2 MB); 4 (samba, nci, webster, mozilla, 21–51 MB) at 256 KiB only
enwik8	first 10^8 bytes	256 KiB prefix only
≥ 1 SDRBench float32 field	whole field	six EXAALT fields at 256 KiB only
UCI household-power text	whole file	256 KiB prefix only
bit-phase-offset extension	—	all 20 files at 256 KiB

- The positive, null, prior-art, and scope controls all behave as designed.
- No non-prior-art natural file meets the per-file threshold, on the 8 whole Silesia members, the 20 files at 256 KiB, or the phase-offset extension.
- The whole-file runs for four Silesia members and for whole enwik8 / SDRBench / UCI exceed the 8 GiB development machine and were **not executed** (§9, §11).

Verdict: inconclusive with respect to the complete preregistered corpus; a clean negative within the achieved coverage. Strictly, the preregistration's "negative" outcome requires the full whole-file list, which was not completed, so the formal hypothesis outcome is *inconclusive*. Within what was run, the result is an unambiguous **layered negative**: on natural data, exact GF(2) structure is often present (RQ-A) and discoverable (RQ-B), and on **no** tested file — at 256 KiB or whole, at any bit phase — does it reduce the representation (RQ-C) or survive composition (RQ-D). Every 256 KiB result and every whole-file result that could be run agree, so the four missing whole-file runs are the only gap between "inconclusive" and "negative".

9. Discussion

9.1 What is established

- The codec recovers and exploits exact GF(2) linear structure when it is deliberately present: complete round-trip, complete accounting, a composed gain up to +49.95 % that scales with rows, independently verified end to end, and not closed by paq8l or paq8px v216.
- The controls establish that this behaviour is not a false positive of the width $\times$ variant search (0 spurious gains on 40 i.i.d. inputs), that metadata is fully counted (null overhead is exactly the 18-byte passthrough header), and that the pipeline degrades gracefully as structure is corrupted.
- On the natural corpora *actually measured* — 8 whole Silesia members plus 20 files at 256 KiB plus the phase-offset extension — the largest positive composed gain on any file is +0.09 % (on a passthrough container, i.e. header perturbation), and G_{pct} is strongly negative on every file where a real GF(2) relation is found. The redundancy the detector locates in natural data is real but trivial (a constant high bit-plane, ASCII structure) and is already captured by the stock compressors from the raw byte order.

9.2 What is not established

- **Not** that natural data contains no exact algebraic redundancy. We tested two specific relation families (axis-aligned fixed-width GF(2), integer affine on parsed tables), extended once to arbitrary bit phase, on one public benchmark corpus plus one scientific dataset (six correlated fields) plus one telemetry file. A materially different family — non-axis-aligned linear via a searched or learned column permutation, or low-degree polynomial relations — is untested.
- **Not** that the approach can never improve compression. Program-synthesis compression demonstrates a real composed win on a favourable domain (model checkpoints). Our negative concerns blind, prior-free, axis-aligned linear discovery on general byte corpora.
- **Not** that the method is universally inferior to existing compressors: on its target structure it wins decisively, and the never-worse guard makes it, by construction, at most 18 bytes worse than passthrough on any input.

9.3 Why the negative is informative

The mechanism is validated: when exact linear structure exists it is found and turned into a real, composition-surviving size reduction. The informative content is that the *analogous* structure a blind axis-aligned detector locates in natural byte corpora is (i) already modelled by stock compressors and (ii) too cheap-to-describe-relative-to-what-it-recovers to survive its own metadata, let alone composition. Establishing a positive on natural data would require either a corpus with genuine exact cross-field structure that stock compressors miss (the FD/CRC cases show what that looks like, and they are prior art), or a detector family expressive enough to find non-trivial structure at acceptable description cost — the direction future work would have to take, and the one the preregistered kill criterion names.

10. Threats to validity

threat	mitigation
encoder and decoder share a bug	independent shared-nothing decoder + independent accounting re-derivation, in CI and on the 10 MB dickens container
vectorised primitives are wrong	400 randomised bit-I/O trials vs a per-bit reference; 60 reconstruct trials vs a per-column XOR reference; 18 byte-frozen container cases
a container has unaccounted bits	finaliser refuses to emit it; independently re-derived by the second decoder
the downstream comparison is unfair (a codec fails on one side)	matched-codec gain over the intersection; both available-codec sets recorded; a reportable positive needs matched sets
the compressed container does not decode back	the <i>primary</i> decoder's raw → encode → compress → decompress → decode → raw is checked for every stock codec on all 122 records (0 failures); the <i>independent</i> shared-nothing decoder runs the same chain on the 21 CI cases and on the whole 10 MB dickens container
the width × variant min inflates results	representation-change null: 0 spurious gains on 40 i.i.d. inputs; never-worse guard caps the downside at passthrough
the negative is a framing / bit-phase artifact	bounded phase-offset extension on all 20 files: G equals the phase-0 value to the byte
a prefix is passed off as a whole file	separate result phase, separate manifest keys, prefix reason on every row
the corpus was chosen after seeing results	corpus list git-locked before any loader output; every file SHA-256-pinned; a changed hash aborts
numbers were transcribed by hand	all tables generated from results/ledger.json; inline figures carry markers checked by check_paper_numbers.py

11. Limitations

- **Detector scope.** Fixed-width, axis-aligned GF(2) linear relations (widths 8–256), plus an integer-affine variant used only for tabular controls, plus one bounded bit-phase-offset extension. The negative is scoped to these families only; it is not a test of algebraic redundancy in general. The non-aligned-period control shows exact structure at a period not among the tried widths is missed by construction. Non-axis-aligned linear relations (via a searched or learned column permutation) and low-degree polynomial relations are untested.
- **Whole-file coverage — the principal limitation.** 8 of the 12 preregistered whole Silesia members were run whole (≤ 10.2 MB). The four largest members (samba, nci, webster, mozilla, 21–51 MB) and whole enwik8 / SDRBench / UCI exceed the 8 GiB development machine and were run only at 256 KiB. Every 256 KiB result, every whole-file result that could be run, and the offset extension agree, so the *direction* of the finding is consistent — but by the preregistration the whole-file list is **not complete**, so the formal hypothesis outcome is *inconclusive with respect to the full corpus* (§8.4). Completing those runs on a ≈ 32 GiB machine is the one outstanding task.
- **Baseline ceiling.** cmix and nncp were not run (hardware). The planted-GF(2) result with paq8l/paq8px bounds — but does not eliminate — the possibility that a stronger mixer would absorb the planted gap; it says nothing about the natural-corpus negative, which does not depend on baseline strength.
- **Corpus breadth.** Four corpus classes; the scientific class is one molecular-dynamics dataset whose six fields are one simulation (effective $n \approx 1-2$). A broader scientific and telemetry sweep is future work.

- **Statistics.** The measurements are deterministic; there is no random sampling from a defined population, so no inferential confidence interval or p-value is claimed. The file is the unit of analysis and the headline is a per-file maximum `G_pct` against the fixed threshold.

12. Conclusion

We asked whether automatically discovered exact GF(2)/affine relations can provide an additional source of lossless compression that survives full description cost and composition with a strong downstream compressor. We built a fixed-width GF(2) relation codec with exhaustive bit accounting, an independent shared-nothing verifier, and a composed-gain metric, and we preregistered the evaluation. The mechanism is validated on deliberately structured data: it recovers planted GF(2) codes and converts them into a composed size reduction of up to +49.95 % that scales with data size and is not closed by two context-mixing compressors.

For the natural-corpus test, the available 8 GiB machine could run only 8 of the 12 preregistered whole Silesia members; the other four members and whole enwik8 / SDRBench / UCI were run only at 256 KiB. **Strictly by the preregistration the hypothesis outcome is therefore inconclusive with respect to the complete corpus; within the achieved coverage the result is a clean layered negative.** No non-prior-art file — 8 whole Silesia members, 20 files at 256 KiB, and the bit-phase-offset extension — yields a composed gain meeting the threshold; the largest positive `G_pct` on any natural file is +0.09 %, on a passthrough container. The functional-dependency and CRC cases reproduce known techniques.

The mechanism is not novel; the contribution is the accounting discipline, the composed evaluation, the controlled validation, and the scoped empirical finding. This is a precise negative, not a refutation of the idea: it establishes that a blind, fixed-width, axis-aligned GF(2) detector does not expose a composed advantage on the natural corpora tested, it identifies the four whole-file runs that separate "inconclusive" from "negative" for the full preregistered corpus, and it states what a future positive would need — either a corpus with exact cross-field structure that stock compressors miss, or a detector family expressive enough to find non-trivial structure at acceptable description cost.

13. Reproducibility

Frozen state. Everything in this paper corresponds to git tag `v1.1-final`. **Environment used:** Windows 11 (build 10.0.22631), 4 logical cores, 8 GiB RAM; Python 3.13.6; numpy 2.2.6, zstandard 0.25.0, brotli 1.2.0 (the rest of `B` is the Python standard library). Corpora are not committed (licence and size); acquisition is scripted and SHA-256-pinned in `results/corpus_manifest.json`, and a changed hash aborts the run.

Headline reproduction — any 8 GiB machine, ~10 min:

```
python -m pip install -e ".[dev]"
python -m pytest -q                                # 636 tests -> all pass
                                                    # (incl. 21 independent-verifier cases
+
                                                    #   randomised property tests + byte-
equivalence)
python verification/independent_verify.py --self-test # -> SELF-TEST: PASS (all 4 container
kinds)
python experiments/controls/run.py                  # -> CONTROLS: PASS (exit 0)
python scripts/build_ledger.py                      # -> 122 records, 0 failures
python scripts/regen_tables.py && python scripts/make_figures.py
python scripts/check_paper_numbers.py               # -> PAPER-NUMBER CHECK: PASS (20
markers)
python scripts/build_pdf.py                         # -> paper/exact-relation-coding.pdf
```

Expected: controls pass; planted GF(2) shows the +32–50 % composed gains of Table B.2; the checker verifies every inline marker against the ledger.

Natural-corpus reproduction (the headline test) — 8 GiB machine:

```
python experiments/natural/run.py --mode whole --only silesia_xml # 8 GiB ok up to ~10 MB
python experiments/offset/run.py --mode slice                     # offset extension, ~15
min
```

These reproduce exactly the 8 whole Silesia members (Table B.3) and the 20-file offset extension (Table B.5): **0 files meet the threshold; largest natural $G_{\text{pct}} = +0.09\%$.**

Complete preregistered corpus — needs > 8 GiB (≈ 32 GiB) RAM:

```
python scripts/reproduce.py --mode whole --offset-whole
```

This additionally runs `samba`, `nci`, `webster`, `mozilla` whole, whole `enwik8`, whole SDRBench fields, and whole UCI — the whole-file runs the present paper could not execute, and that separate its "inconclusive" from a full "negative".

A re-run on the frozen state reproduces every size, gain, and verdict exactly; only timings and timestamps differ. `docs/preregistration.md` (locked), `docs/audit.md` (implementation audit and corrections C1–C2), `docs/prior_art.md`, `docs/statistics.md`, `docs/kill_criterion_status.md`, and `docs/venue_assessment.md` accompany the manuscript.

14. References

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Appendix A — Figures

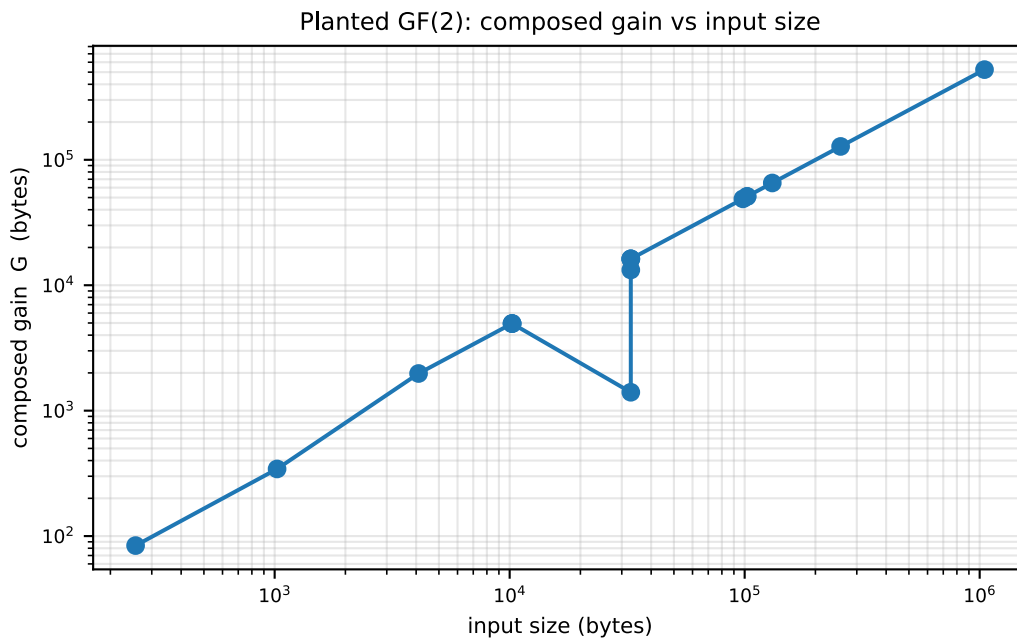


Figure 1. Planted GF(2) codes: composed gain G (bytes) versus input size, log–log. The gain scales approximately with the number of rows because the relation description is a constant of the code. The two low points are a single-flip near-relation control and an affine-parity control.

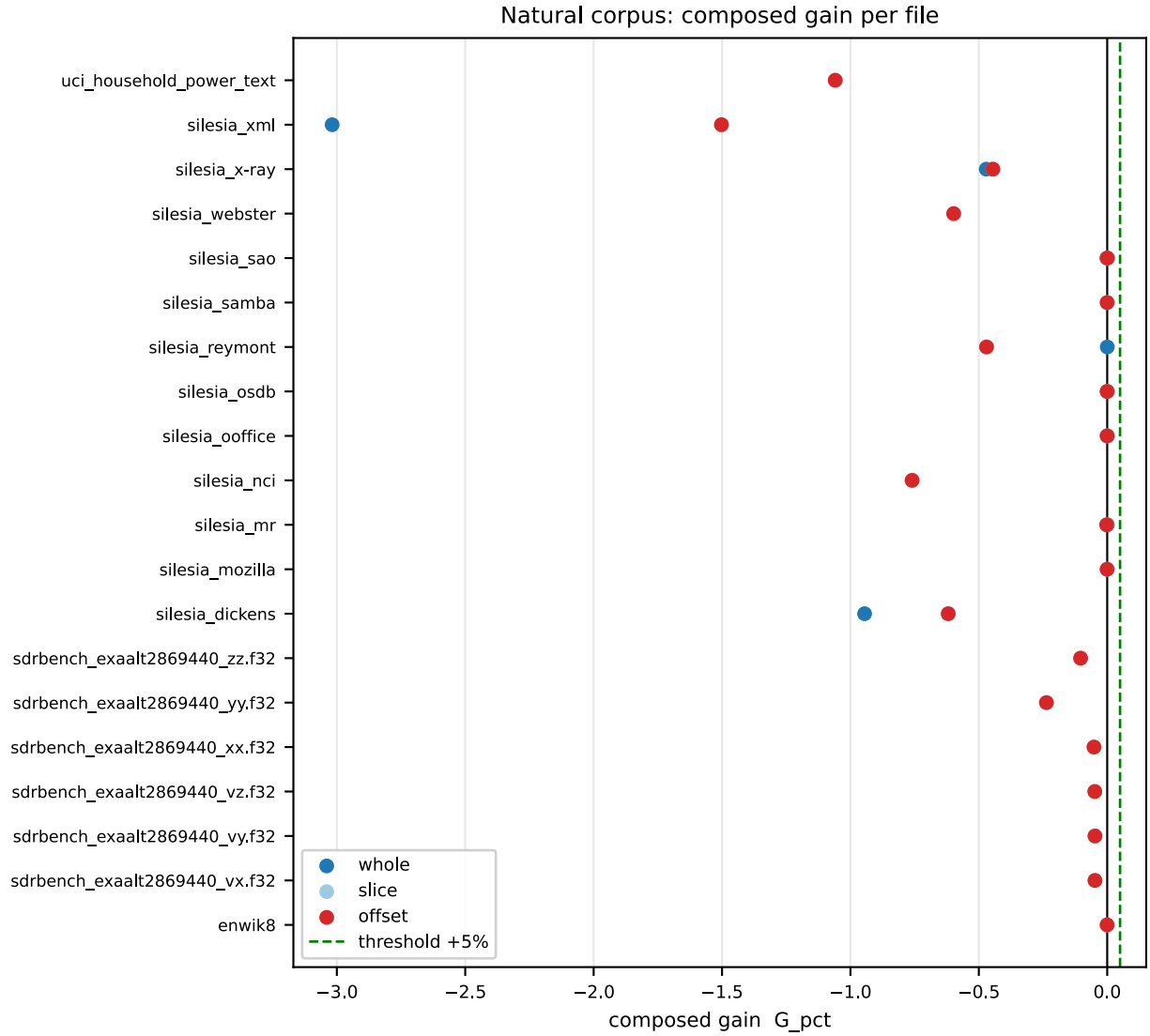


Figure 2. Natural corpus: per-file composed gain G_{pct} for the whole files (blue), the 256 KiB slices (light blue), and the bit-phase-offset extension (red), with the zero line and the preregistered +5 % threshold (dashed). Every point is far below the threshold. Three points sit fractionally to the right of zero — `sao` ($G_{\text{pct}} = +0.0009$), `reymont`, `mr` — all with passthrough containers ($n_{\text{relations}} = 0$); these are sub-0.1 % downstream-compressor reactions to the 18-byte passthrough header, not deduction. Points where a real relation was found are all strongly negative.

Appendix B — Key results tables

The complete auto-generated tables — **B.1** controls, **B.2** planted scaling + context-mixer baseline, **B.3** whole natural, **B.4** 256 KiB natural, **B.5** bit-phase offset, **B.6** prior-art sanity — are in `paper/results_tables.md`, produced from `results/ledger.json` (122 records; 0 accounting, round-trip, or composed round-trip failures). Every row traces to a JSON record under `results/` carrying its dataset hash, git commit, configuration, compressor versions, and verification status. The control numbers (B.1) appear in §6 and the prior-art numbers (B.6) in §8.3; the remaining four tables are reproduced below.

B.2 (extract) — planted GF(2): the mechanism scales

input bytes	relations	container	raw_best	G (bytes)	G_pct	relation bits
256	8	172	260	+84	+32.31 %	80
4,096	16	2,118	4,100	+1,978	+48.24 %	288
32,768	32	16,554	32,772	+16,214	+49.48 %	1,088
102,400	32	51,371	102,405	+51,030	+49.83 %	1,088
256,000	32	128,171	256,005	+127,829	+49.93 %	1,088
1,048,576	64	524,850	1,048,581	+523,726	+49.95 %	4,224

B.3 — whole natural files (8 of 12 Silesia members)

member	bytes	kind	rels	raw_best	container	G (bytes)	G_pct	round-trip
xml	5,345,280	GF2	21	430,390	4,907,484	-1,299,038	-301.83 %	✓
ooffice	6,152,192	passthrough	0	2,426,816	6,152,210	-188	≈ 0	✓
reymont	6,627,202	passthrough	0	1,246,230	6,627,220	+217	≈ 0	✓
sao	7,251,944	passthrough	0	4,415,072	7,251,962	-6,408	-0.15 %	✓
x-ray	8,474,240	GF2	58	4,051,112	6,555,798	-1,907,747	-47.09 %	✓
mr	9,970,564	passthrough	0	2,441,280	9,970,582	+857	≈ 0	✓
osdb	10,085,684	passthrough	0	2,802,792	10,085,702	-25	≈ 0	✓
dickens	10,192,446	GF2	25	2,799,520	9,197,882	-2,645,016	-94.48 %	✓

B.4 — 256 KiB natural files (all 20; phase-o detector). Every input is 262,144 bytes. "pass" = passthrough container (`n_relations = 0`).

file	kind	rels	raw_best	G (bytes)	G_pct
exaalt/vx.f32	GF2	3	226,805	-10,819	-4.8 %
exaalt/vy.f32	GF2	3	226,808	-10,816	-4.8 %
exaalt/vz.f32	GF2	3	226,655	-10,969	-4.8 %
exaalt/xx.f32	GF2	3	224,585	-11,576	-5.2 %
exaalt/yy.f32	GF2	29	187,128	-44,290	-23.7 %
exaalt/zz.f32	GF2	3	215,238	-22,336	-10.4 %
silesia_dickens	GF2	31	77,252	-47,827	-61.9 %
silesia_nci	GF2	1	14,563	-11,065	-76.0 %
silesia_reymont	GF2	1	53,158	-24,988	-47.0 %
silesia_webster	GF2	1	62,139	-37,157	-59.8 %
silesia_x-ray	GF2	58	129,632	-57,729	-44.5 %
silesia_xml	GF2	1	11,559	-17,365	-150.2 %
uci_household_power_text	GF2	60	28,160	-29,828	-105.9 %
enwik8	pass	0	73,306	-39	≈ 0
silesia_mozilla	pass	0	111,811	-54	≈ 0
silesia_mr	pass	0	55,898	-119	≈ 0
silesia_ooffice	pass	0	123,528	-16	≈ 0
silesia_osdb	pass	0	82,475	-19	≈ 0
silesia_samba	pass	0	153,153	-45	≈ 0
silesia_sao	pass	0	167,536	+148	+0.09 %

Every GF(2) container is strongly composition-negative; every passthrough container is within header noise. Round-trip and composed round-trip hold for all 20 (Table in `results_tables.md`).

B.5 (summary) — bit-phase-offset extension (20 files, 256 KiB). For every file the offset-search `G` equals the corresponding phase-0 `G` in the table above **to the byte**; 0 files cross the +5 % threshold; 0 files improve on phase 0 by more than header perturbation. The fixed-width axis-aligned negative is robust to bit phase.